MOTHERBOARD

The main circuit board that connects every part of a computer – explained step by step



What we will learn

1 The basics

What a motherboard is, its types, segments and brands

2 Form factors

ATX family and other board sizes and shapes

3 Buses

Data, address and control buses

4 Core parts

CPU socket, memory slots, CMOS battery, expansion slots, chipset

5 Expansion cards

Graphics, audio and network (NIC) cards

6 Ports & cables

Back-panel ports, cables, connectors and a handy command

WHAT IS MOTHERBOARD

FROM THE LECTURE

It is known as the Interconnection of Components in a computer

It is a Circuit board which have ports and sockets with transistors coating to connect components together

It is responsible for contact of flow of data and instruction with the help of BUSES

Two types of Motherboard 1. Plane 2. Mark

WHAT IS MOTHERBOARD – Explained

1 Interconnection

Every part of the computer – CPU, RAM, disk, cards – plugs into the motherboard and can talk to the others through it.

2 Circuit board

A flat board (PCB) with sockets and ports. Thin copper tracks and tiny electronic parts carry the signals.

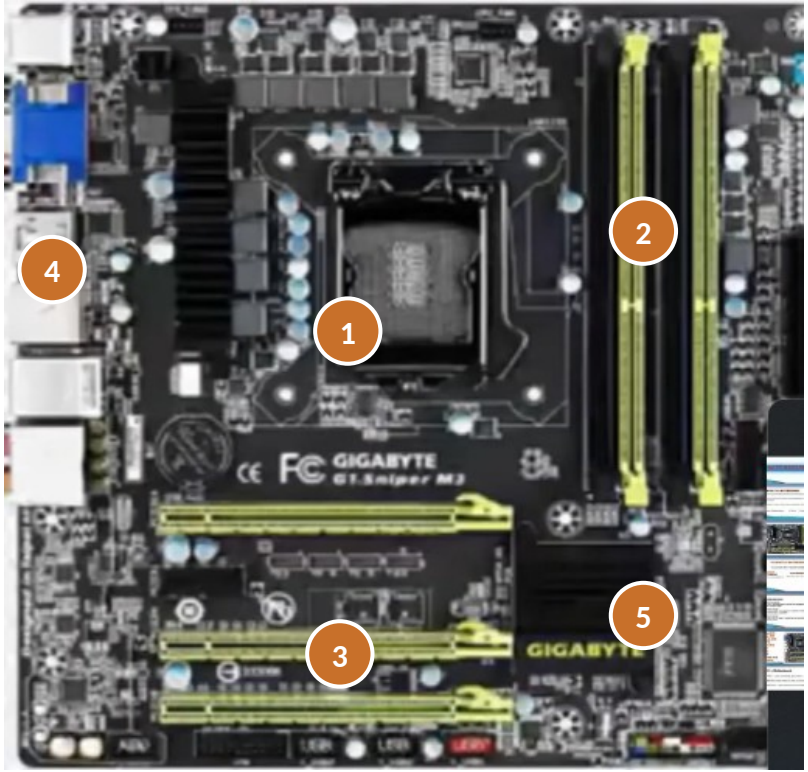
3 Buses

Buses are the wire pathways on the board that carry data and instructions between parts.

4 Plane and Mark

Plane = a plain / generic board. Mark = a branded board from a known maker (our interpretation of the lecture).

Inside a motherboard



Example: Gigabyte G1.Sniper M3 board from the lecture

1. CPU socket
Holds the processor – the brain.

2. Memory (RAM) slots
Hold RAM sticks – the short-term memory.

3. Expansion slots
Long slots for graphics, sound and network cards.

4. Back-panel ports
Connectors for keyboard, mouse, USB, LAN, audio.

5. Chipset area
Controls data flow between CPU, memory and devices.

SEGMENTS OF MOTHERBOARD

CATEGROIZED INTO 2 DIFFERENT SEGMENTS

1 INTEGRATED MOTHERBOARD

COMES WITH ALL PARTS INTEGRATED WITH IT

Explained

Graphics, sound and network are built into the board. Cheaper, compact and ready to use – but hard to upgrade those parts.

2 NON-INTEGRATED MBOARD

ADD SOME OTHER PARTS OR PARTS AS PER REQUIREMENT.

Explained

You add separate cards (graphics, sound, LAN) as needed. Costs more, but gives better performance and easy upgrades.

MAJOR BRANDS

1

DESKTOP MBOARD

It is used in Desktop Systems. It comes with Upgradation Capability. We can Upgrade it Easily

Full-size board in a tower case; easy to swap CPU, RAM and cards.

2

LAPTOP MBOARD

It is used in Laptop. All components are placed in limited space. Rare mboard allows Upgradation

Compact, custom-shaped for each model; parts are often soldered, so upgrades are limited.

3

SERVER MBOARD

It is used in Server. Bigger in size having many ports and designed for high end works

Many CPU/RAM slots and ports, built to run 24×7 for heavy workloads.

Motherboard form factors

1.Mini ATX
2.Flex ATX
3.Micro ATX
4.EATX
5.Pico ATX
6.ITX
7.NLX
8.BTX
9.Mobile ATX
10.WATX



Form factors – Explained

1. Mini ATX A slightly smaller version of the ATX board.

2. Flex ATX Reduced ATX for small, low-cost cases.

3. Micro ATX Compact (about 244 × 244 mm) with fewer slots.

4. EATX Extended ATX: bigger, more slots – servers.

5. Pico ATX Very small board for tiny, low-power PCs.

6. ITX Small, low-power boards for compact PCs.

7. NLX Low-profile board for slim desktop cases.

8. BTX Intel design (2004) for better cooling.

9. Mobile ATX Smaller ATX-style board for mobile use.

10. WATX Workstation ATX: large, for workstations.

BUSES in Motherboard

DATA BUS

It Sends and Receives data to Memory.
Two-way road: carries the actual data between CPU, memory and devices.

ADDRESS BUS

It Sends Address of Inputs to the Memory or RAM
One-way road: tells memory WHERE to read or write (like a house number).

CONTROL BUS

It Makes the Communication between CPU and Components
Carries signals such as read / write and timing, so all parts work together.

CPU SOCKET

FROM THE LECTURE

A CPU socket or CPU slot is a mechanical component(s) that provides mechanical and electrical connections between a microprocessor and a printed circuit board (PCB). This allows the CPU to be replaced without soldering.



CPU SOCKET – Explained

1 Mechanical connection

The socket holds the processor firmly; a lever clamps it down.

2 Electrical connection

Hundreds of pins or contacts link the CPU to the rest of the board.

3 No soldering needed

You can remove and replace the CPU easily – this makes upgrades possible.

4 Match is important

Each CPU needs the right socket type, so check compatibility before buying.

MEMORY SLOTS

FROM THE LECTURE

A memory slot, memory socket, or RAM slot is what allows computer memory (RAM) to be inserted into the computer. Depending on the motherboard, there will usually be 2 to 4 memory slots (sometimes more on high-end motherboards) and are what determine the type of RAM used with the computer. The most common types of RAM are SDRAM and DDR for desktop computers and SODIMM for laptop computers, each having various types and speeds. In the below picture, is an example of what memory slots may look like inside a desktop computer. In this picture, there are three open available slots for three memory sticks.



MEMORY SLOTS – Explained

1 Where RAM goes

RAM sticks click into these slots next to the CPU socket.

2 How many

Usually 2 to 4 slots; high-end boards can have more.

3 Type of RAM

The slot decides the RAM type: SDRAM / DDR for desktops, SODIMM for laptops.

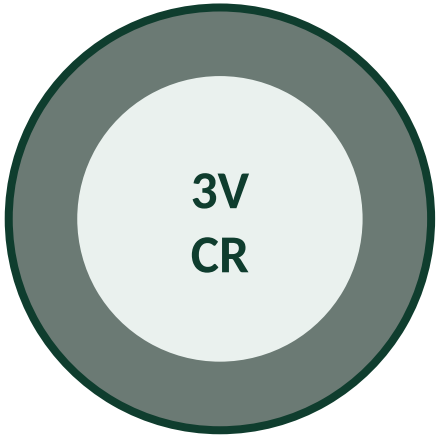
4 In the picture

Three open slots are visible, so three memory sticks can be added.

CMOS BATTERY

FROM THE LECTURE

Nonvolatile BIOS memory refers to a small memory on PC motherboards that is used to store BIOS settings. It was traditionally called CMOS RAM because it used a volatile, low-power complementary metal-oxide-semiconductor (CMOS) SRAM (such as Motorola MC146818 or similar) powered by a small battery when system power was off.



Coin-cell CMOS battery

CMOS BATTERY – Explained

1 Stores BIOS settings

Date, time and boot order are kept in this small memory.

2 Needs power

CMOS SRAM loses data without power, so a small coin battery feeds it when the PC is off.

3 Nonvolatile feel

Because of the battery, the settings survive switch-off – hence "nonvolatile BIOS memory".

4 Signs it is weak

Wrong date or time and lost BIOS settings mean the battery needs replacing.

ISA. PCI & AGP SLOTS

FROM THE LECTURE

ISA, or Industry Standard Architecture, is an 8bit or 16bit parallel bus system that allowed up to 6 devices to be connected to a PC. Virtually all IBM-compatible PCs made before the Pentium were based on the ISA (IBM's PC AT) bus. This asynchronous bus architecture uses 16-bit addresses and an 8-MHz clock and handles a maximum data throughput of 2 MB/s to 3 MB/s.

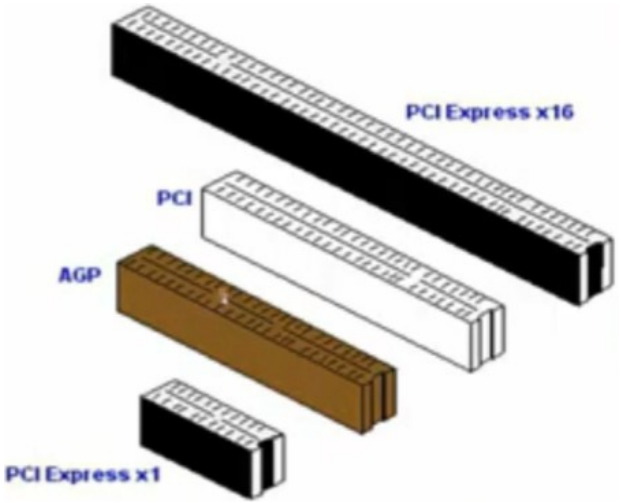


ISA, PCI and AGP

FROM THE LECTURE

AGP, Accelerates Graphics Port is a high-speed point-to-point channel for attaching a video card to a computer's motherboard. The primary advantage of AGP over PCI is that it provides a dedicated pathway between the slot and the processor rather than sharing the PCI bus.

PCI, Peripheral Component Interconnect bus uses a local bus system; this system is independent of the processor bus speed. The PCI architecture incorporates its own chip set which link the local bus to the main bus, these links are called bridges, there are two bridges which "north and south bridge"



ISA, PCI and AGP – Explained

1 ISA (oldest)

Slow 8/16-bit bus from IBM PC AT days; only about 2–3 MB/s. Now obsolete.

2 PCI

Own local bus, independent of CPU speed. Linked to the main bus by bridges (north and south).

3 AGP

Dedicated fast path just for the video card, so graphics do not share the PCI bus.

4 PCI Express

Newer slots shown in the picture (x1 and x16). They replaced PCI and AGP in modern boards.

CHIPSET

FROM THE LECTURE

A chipset is a set of electronic components in an integrated circuit that manages the data flow between the processor, memory and peripherals. It is usually found in the motherboard of a computer.

Chipsets are usually designed to work with a specific family of microprocessors. Because it controls communications between the processor and external devices, the chipset plays a crucial role in determining system performance.



CHIPSET – Explained

1 Traffic controller

Manages data flow between processor, memory and peripherals.

2 On the motherboard

Usually sits on the board, often under a small heatsink.

3 Made for a CPU family

Each chipset works with a specific family of processors (Intel P67 is shown).

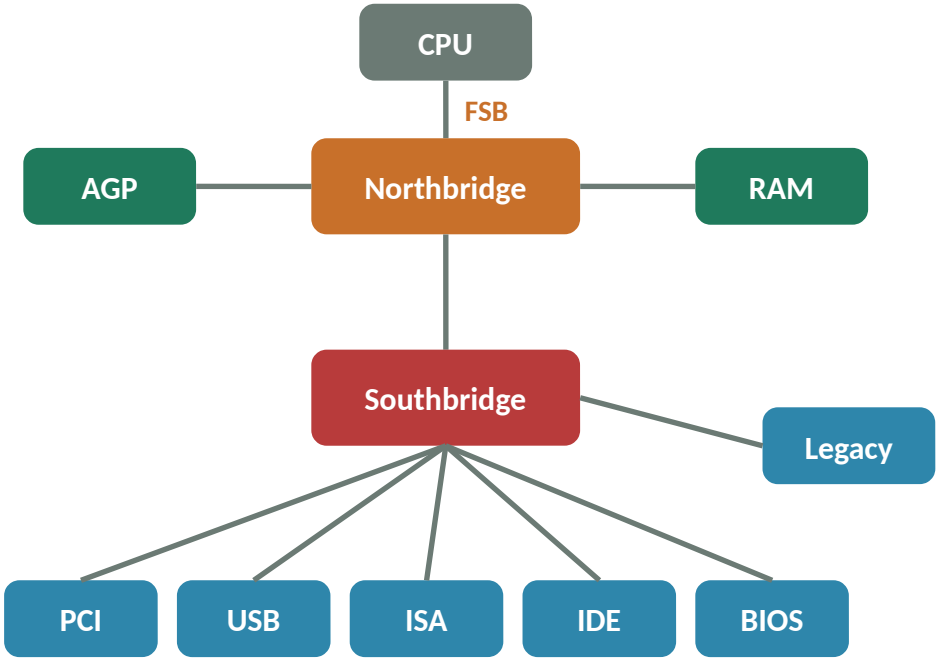
4 Decides performance

All talk between CPU and outside devices passes through it, so it affects speed.

SOUTHBRIDGE

FROM THE LECTURE

The southbridge is one of the two chips in the core logic chipset on a personal computer (PC) motherboard, the other being the northbridge. The southbridge typically implements the slower capabilities of the motherboard in a northbridge/southbridge chipset computer architecture.



Northbridge and Southbridge – Explained

1 Northbridge

Fast side: links the CPU (through the Front Side Bus, FSB) to RAM and the graphics (AGP) slot.

2 Southbridge

Slower side: handles PCI slots, USB, ISA, IDE drives, BIOS and legacy ports.

3 Two-chip chipset

Together they form the core logic chipset; the northbridge talks to the southbridge.

4 Why split?

Fast parts need a quick, direct path; slow devices are managed separately.

GRAPHICS card

FROM THE LECTURE

A **Graphics Card** is a piece of computer hardware that produces the image you see on a monitor. **The Graphics Card** is responsible for rendering an image to your monitor, **it does** this by converting data into a signal your monitor can understand.



GRAPHICS card – Explained

1 Makes the picture

Everything on screen – desktop, video, games – is produced by this card.

2 Renders images

Draws each frame from raw data.

3 Signal for monitor

Converts data into a signal the monitor understands (cable outputs on the card).

4 Where it plugs

In an AGP or PCI Express slot on the motherboard.

audio card

FROM THE LECTURE

A sound **card** is an expansion **card** or IC for producing sound on a computer that can be heard through speakers or headphones. Although the computer **does** not need a sound device to function, they are included on every machine in one form or another, either in an expansion slot or built into the motherboard



audio card – Explained

1 Produces sound

Turns digital audio into sound for speakers or headphones.

2 Expansion card or IC

Can be a separate card or a chip on the board.

3 Not essential

The PC works without it, but every machine has one in some form.

4 Colour jacks

The coloured sockets on the card are the audio connectors.

Nic card

FROM THE LECTURE

A **network** interface **card** (**NIC**) is a hardware component without which a computer cannot be connected over a **network**. It is a circuit board installed in a computer that provides a dedicated **network** connection to the computer. It is also called **network** interface controller, **network** adapter or LAN adapter.



Nic card – Explained

1 Needed for networking

Without it a computer cannot join a network.

2 Dedicated connection

Gives the PC its own network link.

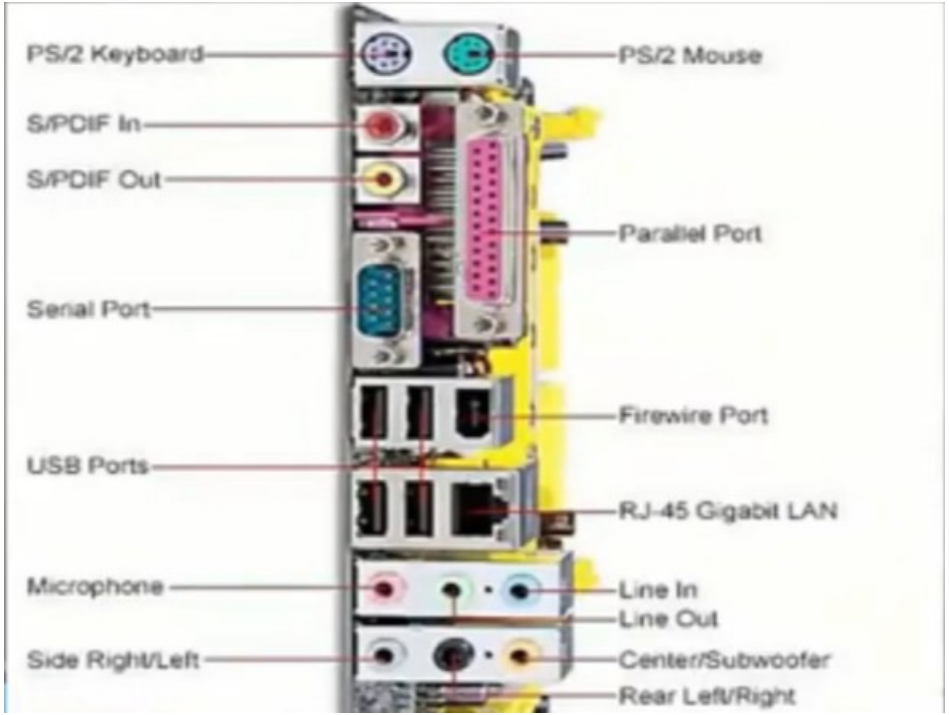
3 Other names

Network interface controller, network adapter, LAN adapter.

4 In the picture

A server-type card with two fibre ports and a heatsink.

BACK PANEL CONNECTORS AND PORTS



Keyboard & mouse: PS/2 ports

Audio: S/PDIF In/Out, Microphone, Line In/Out, speakers

Old ports: Serial, Parallel

Fast data: USB, FireWire

Network: RJ-45 Gigabit LAN

PORTS

1. PS2 MOUSE
2. LAN
3. AUDIO PORT
4. PS2 KEYBOARD
5. USB 3.0
6. SERIAL
7. VGA
8. USB 2.0



Tip: PS/2 ports connect old keyboards and mice, LAN is for the network cable, audio jacks are colour-coded, USB 3.0 is faster than USB 2.0, and VGA/serial are older connectors.

CABLES & CONNECTORS

1. SATA & IDE CABLES
2. POWER CONNECTION WIRES
3. CPU FAN WIRE
4. MOLEX WIRE

1. SATA & IDE PORT 7 & 40 PINS
2. 20-24 PINS ATX PORT
3. 4 PINS BRY PORT
4. FOR POWER IN HDD 4 PINS






Cables & connectors – Explained

1 SATA & IDE cables

Carry data between storage drives and the board (SATA 7 pins, IDE 40 pins).

2 Power connection wires

The 20-24 pin ATX connector brings power from the supply to the board.

3 CPU fan wire

The small 4-pin connector that powers and controls the CPU fan.

4 Molex wire

Older 4-pin plug that supplies power to drives such as an HDD.

Check your motherboard details

```
wmic baseboard get product, manufacturer, version, serialnumber
```

1 Open CMD

Press Windows + R, type cmd and press Enter.

2 Run the command

Type the line above exactly and press Enter.

3 Read the result

It shows the board model (product), maker, version and serial number.

Thank you

Motherboard = the board that interconnects all computer components using buses, sockets, slots and ports.

